Q1] CREATE DATABASE Restaurant;
USE Restaurant;

```
CREATE TABLE Category(
category_id INT,
category_name VARCHAR(50)
);


CREATE TABLE Dish(
dish_id INT,
dish_name VARCHAR(50),
category_id INT,
price DECIMAL(10,2),
dish_rank VARCHAR(50)
);

	CREATE TABLE Customer(
customer_id INT,
customer_name VARCHAR(50),
phone VARCHAR(50)
);

	CREATE TABLE Orders(
order_id INT,
customer_id INT,
order_date DATE
);

	CREATE TABLE OrderSpecs(
specs_id INT,
	order_id INT,
	dish_id INT,
	quantity INT
);
```

Q2] ALTER TABLE Category ADD PRIMARY KEY(category_id);
ALTER TABLE Dish ADD PRIMARY KEY(dish_id);
ALTER TABLE Customer ADD PRIMARY KEY(customer_id);
ALTER TABLE Orders ADD PRIMARY KEY(order_id);
ALTER TABLE OrderSpecs ADD PRIMARY KEY(specs_id);

Q3] INSERT INTO Category Values
(1,"Chinese"),(2,"Indian"),(3,"Punjabi"),(4,"Italian"),(5,"Afgani");

INSERT INTO Dish Values

```
(101,"Veg Noodles",1,160.00,"Regular"),
(102,"Veg Special Thali",1,200.00,"Popular"),
      (103,"Dal Makhani",2,180.00,"Special"),
      (104,"Pasta",3,360.00,"Popular"),
(105,"Chicken Afgani",4,260.00,"Regular");


INSERT INTO Customer Values
(1,"Sai","9090796659"),
      (2,"Raj","9093079659"),
      (3,"Neha","9019079659"),
(4,"Amit","9209079659"),
            (5,"Rahul","9009079659"),




      INSERT INTO Orders Values
(201,1,"2026-10-11"),
(202,2,"2026-10-12"),
(203,3,"2026-10-13"),
(204,4,"2026-10-14"),
(205,5,"2026-10-15");


      INSERT INTO OrderSpecs Values
      (1,201,101,2),
            (2,202,102,1),
            (3,203,103,3),
            (4,204,104,1),
            (5,205,105,2);




Q4]    ALTER TABLE Dish
ADD FOREIGN KEY(category_id) REFERENCES Category(category_id);

ALTER TABLE Orders
ADD FOREIGN KEY(customer_id) REFERENCES Customer(customer_id);

ALTER TABLE OrderSpecs
ADD FOREIGN KEY(order_id) REFERENCES Orders(order_id)




Q5]   SELECT * FROM Dish
    WHERE price < 250 OR category_id=4;
```

Q6] SELECT dish_name,category_id,price,
RANK() OVER(ORDER BY price DESC) as dish_rank
FROM Dish;

Q7] SELECT category_id, COUNT(*) AS total_dishes
FROM Dish
GROUP BY category_id

Q8] SELECT dish_name,price
FROM Dish
WHERE price=(SELECT MAX(price) FROM Dish);

Q9] CREATE VIEW DishView AS
SELECT dish_id,dish_name,price
FROM Dish;

SELECT * FROM DishView;

UPDATE DishView
SET price=240.00
WHERE dish_id=101;

SELECT * FROM Dish
WHERE dish_id=101;

Q10] SELECT d.dish_name,c.category_name
FROM Dish d
LEFT JOIN Category c ON
d.category_id=c.category_id;

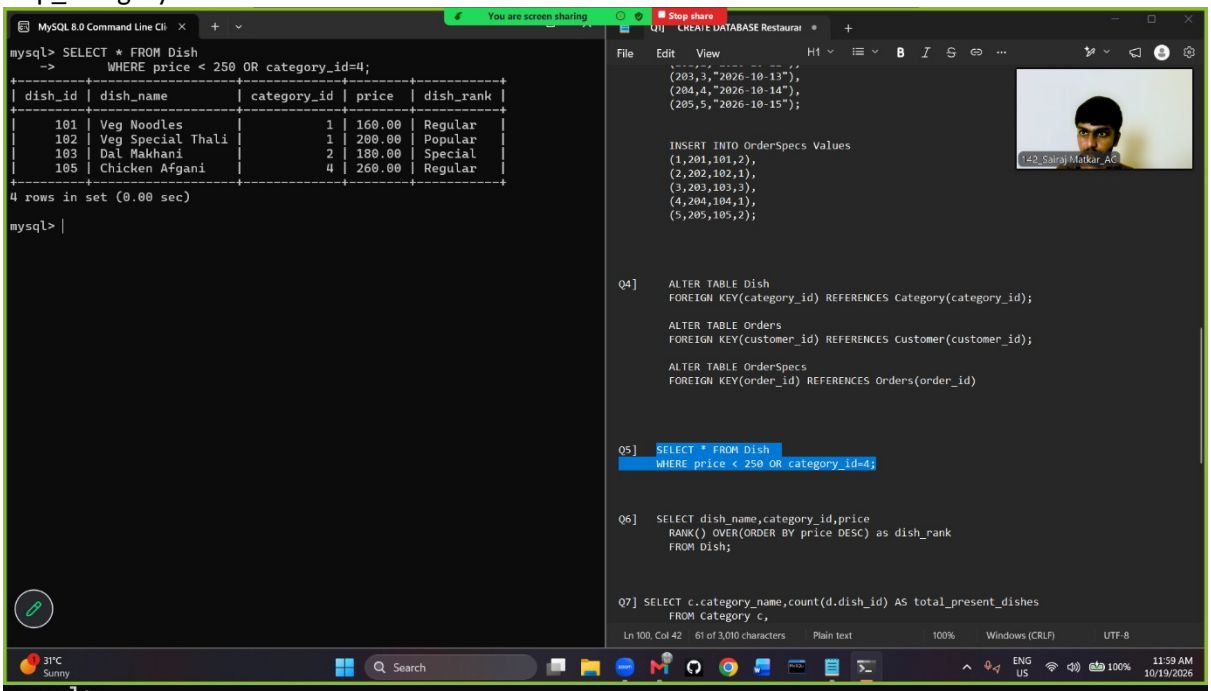
Q11] CREATE TEMPORARY TABLE ChineseMenu AS
SELECT dish_id, dish_name, price
FROM Dish
WHERE category_id = 1;

SELECT * FROM ChineseMenu;

DROP TEMPORARY TABLE ChineseMenu;

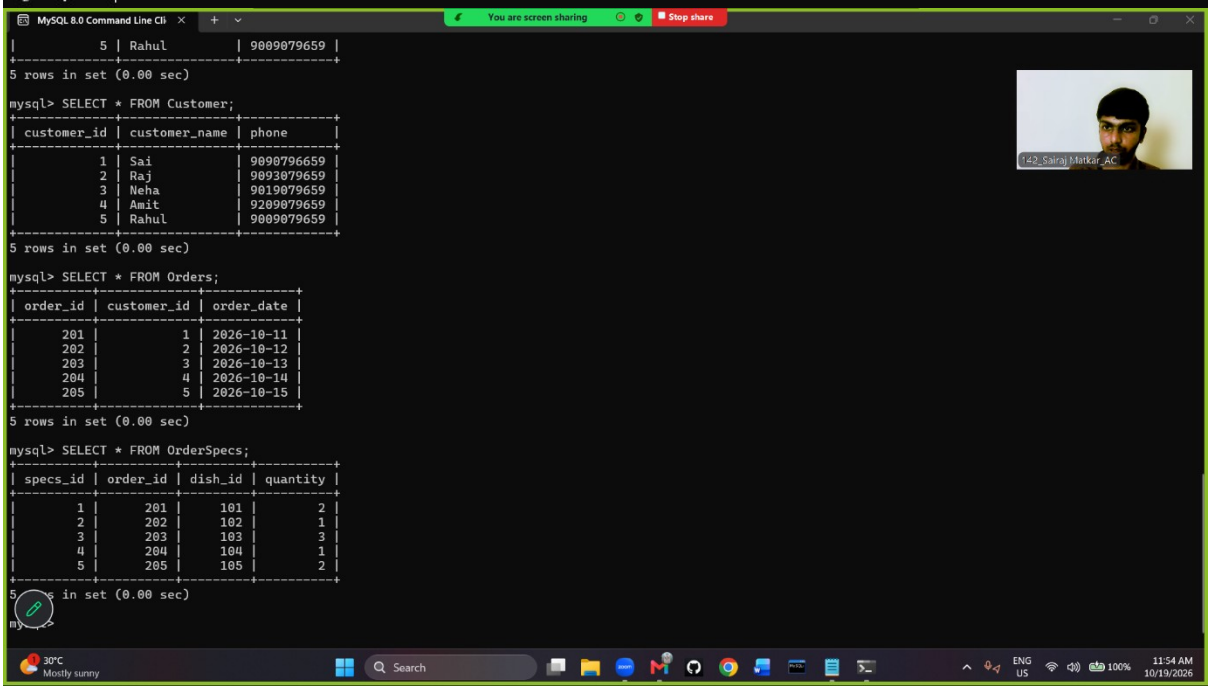
Q12] CREATE PROCEDURE CountDishesByCat(

IN p_category




```
mysql>
mysql>      SELECT * FROM Dish
    ->      WHERE dish_id=101;
+----------+--------------+--------------+--------+-----------+
| dish_id  | dish_name    | category_id  | price  | dish_rank |
+----------+--------------+--------------+--------+-----------+
|      101 | Veg Noodles  |            1 | 240.00 | Regular   |
+----------+--------------+--------------+--------+-----------+
1 row in set (0.00 sec)

mysql>
```



```
    ->          (2,202,102,1),
    ->          (3,203,103,3),
    ->          (4,204,104,1),
    ->          (5,205,105,2);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM CATEGORY;
+-------------+---------------+
| category_id | category_name |
+-------------+---------------+
|           1 | Chinese       |
|           2 | Indian        |
|           3 | Punjabi       |
|           4 | Italian       |
|           5 | Afgani        |
+-------------+---------------+
5 rows in set (0.01 sec)

mysql> SELECT * FROM Dish;
+---------+-------------------+-------------+--------+-----------+
| dish_id | dish_name         | category_id | price  | dish_rank |
+---------+-------------------+-------------+--------+-----------+
|     101 | Veg Noodles       |           1 | 160.00 | Regular   |
|     102 | Veg Special Thali |           1 | 200.00 | Popular   |
|     103 | Dal Makhani       |           2 | 180.00 | Special   |
|     104 | Pasta             |           3 | 360.00 | Popular   |
|     105 | Chicken Afgani    |           4 | 260.00 | Regular   |
+---------+-------------------+-------------+--------+-----------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM Customer;
+-------------+---------------+------------+
| customer_id | customer_name | phone      |
+-------------+---------------+------------+
|           1 | Sai           | 9090796659 |
|           2 | Raj           | 9093079659 |
|           3 | Neha          | 9019079659 |
|           4 | Amit          | 9209079659 |
|           5 | Rahul         | 9009079659 |
+-------------+---------------+------------+
```

```
mysql> SELECT dish_name,price
    ->      FROM Dish
    ->      WHERE price=(SELECT MAX(price) FROM Dish);
+-----------+--------+
| dish_name | price  |
+-----------+--------+
| Pasta     | 360.00 |
+-----------+--------+
1 row in set (0.00 sec)

mysql>
```

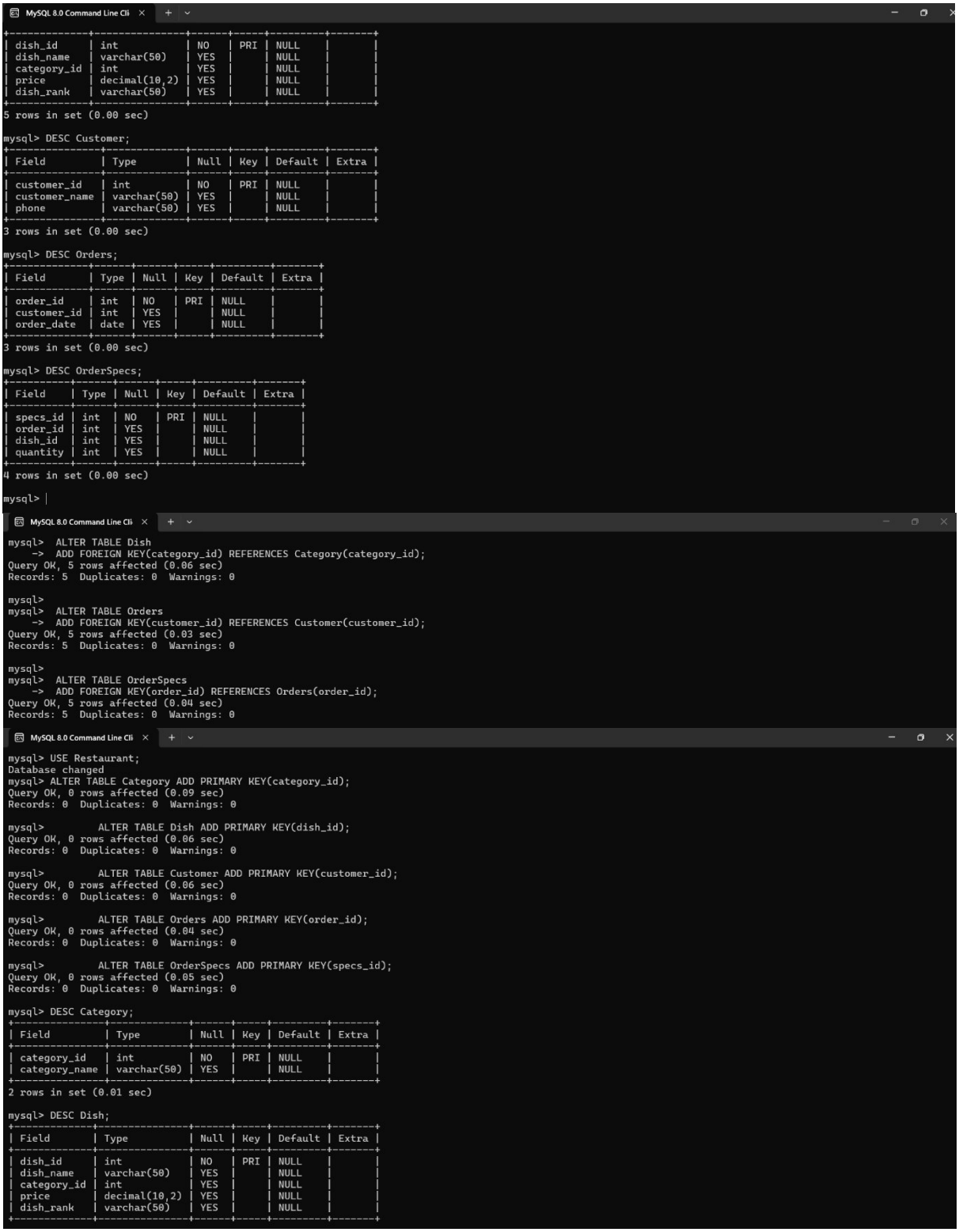
```
    ->  customer_name VARCHAR(50),
    ->  phone VARCHAR(50),
    ->  );
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near ')
' at line 5
mysql>   CREATE TABLE Customer(
    ->  customer_id INT,
    ->  customer_name VARCHAR(50),
    ->  phone VARCHAR(50)
    ->  );
Query OK, 0 rows affected (0.02 sec)

mysql>   CREATE TABLE Orders(
    ->  order_id INT,
    ->  customer_id INT,
    ->  order_date DATE
    ->  );
Query OK, 0 rows affected (0.02 sec)

mysql>          CREATE TABLE OrderSpecs(
    ->  specs_id INT,
    ->          order_id INT,
    ->          dish_id INT,
    ->          quantity INT
    ->  );
Query OK, 0 rows affected (0.02 sec)

mysql> SHOW TABLES;
+----------------------+
| Tables_in_restaurant |
+----------------------+
| category             |
| customer             |
| dish                 |
| orders               |
| orderspecs           |
+----------------------+
5 rows in set (0.02 sec)

m
```



```
MySQL 8.0 Command Line Cli

+-------------+---------------+------+-----+---------+-------+
| dish_id     | int           | NO   | PRI | NULL    |       |
| dish_name   | varchar(50)   | YES  |     | NULL    |       |
| category_id | int           | YES  |     | NULL    |       |
| price       | decimal(10,2) | YES  |     | NULL    |       |
| dish_rank   | varchar(50)   | YES  |     | NULL    |       |
+-------------+---------------+------+-----+---------+-------+
5 rows in set (0.00 sec)

mysql> DESC Customer;
+---------------+-------------+------+-----+---------+-------+
| Field         | Type        | Null | Key | Default | Extra |
+---------------+-------------+------+-----+---------+-------+
| customer_id   | int         | NO   | PRI | NULL    |       |
| customer_name | varchar(50) | YES  |     | NULL    |       |
| phone         | varchar(50) | YES  |     | NULL    |       |
+---------------+-------------+------+-----+---------+-------+
3 rows in set (0.00 sec)

mysql> DESC Orders;
+-------------+------+------+-----+---------+-------+
| Field       | Type | Null | Key | Default | Extra |
+-------------+------+------+-----+---------+-------+
| order_id    | int  | NO   | PRI | NULL    |       |
| customer_id | int  | YES  |     | NULL    |       |
| order_date  | date | YES  |     | NULL    |       |
+-------------+------+------+-----+---------+-------+
3 rows in set (0.00 sec)

mysql> DESC OrderSpecs;
+----------+------+------+-----+---------+-------+
| Field    | Type | Null | Key | Default | Extra |
+----------+------+------+-----+---------+-------+
| specs_id | int  | NO   | PRI | NULL    |       |
| order_id | int  | YES  |     | NULL    |       |
| dish_id  | int  | YES  |     | NULL    |       |
| quantity | int  | YES  |     | NULL    |       |
+----------+------+------+-----+---------+-------+
4 rows in set (0.00 sec)

mysql>
```

```
MySQL 8.0 Command Line Cli

mysql>  ALTER TABLE Dish
    ->  ADD FOREIGN KEY(category_id) REFERENCES Category(category_id);
Query OK, 5 rows affected (0.06 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql>
mysql>  ALTER TABLE Orders
    ->  ADD FOREIGN KEY(customer_id) REFERENCES Customer(customer_id);
Query OK, 5 rows affected (0.03 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql>
mysql>  ALTER TABLE OrderSpecs
    ->  ADD FOREIGN KEY(order_id) REFERENCES Orders(order_id);
Query OK, 5 rows affected (0.04 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
MySQL 8.0 Command Line Cli

mysql> USE Restaurant;
Database changed
mysql> ALTER TABLE Category ADD PRIMARY KEY(category_id);
Query OK, 0 rows affected (0.09 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql>         ALTER TABLE Dish ADD PRIMARY KEY(dish_id);
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql>         ALTER TABLE Customer ADD PRIMARY KEY(customer_id);
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql>         ALTER TABLE Orders ADD PRIMARY KEY(order_id);
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql>         ALTER TABLE OrderSpecs ADD PRIMARY KEY(specs_id);
Query OK, 0 rows affected (0.05 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> DESC Category;
+---------------+-------------+------+-----+---------+-------+
| Field         | Type        | Null | Key | Default | Extra |
+---------------+-------------+------+-----+---------+-------+
| category_id   | int         | NO   | PRI | NULL    |       |
| category_name | varchar(50) | YES  |     | NULL    |       |
+---------------+-------------+------+-----+---------+-------+
2 rows in set (0.01 sec)

mysql> DESC Dish;
+-------------+---------------+------+-----+---------+-------+
| Field       | Type          | Null | Key | Default | Extra |
+-------------+---------------+------+-----+---------+-------+
| dish_id     | int           | NO   | PRI | NULL    |       |
| dish_name   | varchar(50)   | YES  |     | NULL    |       |
| category_id | int           | YES  |     | NULL    |       |
| price       | decimal(10,2) | YES  |     | NULL    |       |
| dish_rank   | varchar(50)   | YES  |     | NULL    |       |
+-------------+---------------+------+-----+---------+-------+
```